Learning Center Probability Map for Detecting Objects in Aerial Images

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***Abstract*— One fundamental problem in Earth Vision is to accurately find the locations and identify the categories of the interesting objects in the aerial images, for which oriented bounding boxes (OBBs) are usually employed to depict better the objects emerging with arbitrary orientations. However, the regression of the OBBs always suffers from the ambiguous problem in the definition of the regression targets, which often reduces the convergency efficiency and decreases the detection accuracy. Although there are some methods like the binary segmentation map that can handle this problem, it brings a new problem of ambiguous background pixels in the OBBs. In this article, we propose to cast the OBB regression as a center-probability-map (CenterMap)-prediction problem, thus largely eliminating the ambiguities on the target definitions and the background pixels. The predicted CenterMaps are then used to generate the OBBs. The CenterMap OBB representation is simple, yet effective. Furthermore, to distinguish better the interesting objects from the cluttered background, a weighted pseudosegmentation-guided attention network is adopted to provide the object-level features for predicting the horizontal bounding boxes and the OBBs. The experimental results on three widely used data sets, i.e., DOTA, HRSC2016, and UCAS-AOD, demonstrate the effectiveness of our proposed method.**

***Index Terms*— Aerial images, center probability map (CenterMap), object detection, oriented bounding boxes (OBBs).**

I. Introduction

AUTOMATIC object detection in aerial images has benefited a wide variety of practical applications, e.g., location-based services, large-scale surveillance, and intelligent transportation. Although recent investigation for the object-detection problem has already achieved a great progress, it is still a very challenging problem when aerial images are involved due to arbitrary orientation, cluttered background, etc.

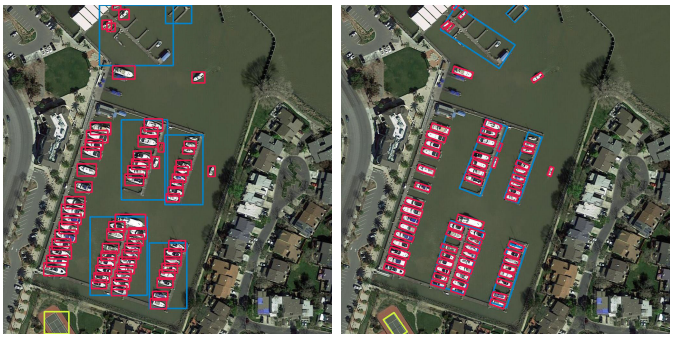


Fig. 1. Examples of (Left) HBB- and (Right) OOB-detection results produced with the proposed network on the DOTA data set. *Red, blue, and yellow* bounding boxes mean *ship*, *harbor*, and *tennis court* classes, respectively. Best viewed in color and zoomed-in view.

The object-detection methods for general purpose are usually based on the horizontal bounding box (HBB), which has achieved a great success in natural scenes [1]–[8]. Recently, the HBB-based methods are also extensively used for object detection in the aerial images [9]–[12]. However, the commonly used HBB cannot accurately describe a majority of objects in the aerial images, because the objects are in arbitrary orientation and often densely packed in the aerial images, as shown in Fig. 1. To address this challenge, some data sets [13]–[16] adopted oriented bounding box (OBB) instead of HBB to enclose compactly the oriented objects by annotating the objects in the aerial images. The OBBs can not only compactly enclose the oriented objects but also retain the orientation information that is very useful for further processing. For instance, as shown in Fig. 1, one HBB might contain more than one object due to the properties of the densely packed and arbitrary orientation. Nevertheless, one OBB normally contains one object, because OBB can compactly enclose the object in the aerial images.

The above data sets bring a new problem to detect the oriented objects in the aerial images. A branch of works [15], [17]–[21] treat this problem as an equivalent regression task. However, they fail to address the inherent order ambiguity in defining the regression targets. For example, the OBB that is represented as four vertices $\{V_i = (x_i, y_i)|i = 1, 2, 3, 4\}$ [15]

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has four different representative ways by changing the order of vertices, i.e., although $\{V_1, V_2, V_3, V_4\}$, $\{V_2, V_3, V_4, V_1\}$, $\{V_3, V_4, V_1, V_2\}$, and $\{V_4, V_1, V_2, V_3\}$ all represent the same OBB, they are entirely different regression targets for the general regressors. Aiming to obtain a unique representation for the OBB, some tricks such as defining the first vertex by a certain rule are used [15]. However, the problem is still unsolved, because they may allow a similar feature map with obvious different OBB representations of ground truth, which would cause inconsistency. Due to the ambiguity of the regression targets for the regressors, the training becomes hard to converge, and the accuracy of the HBB task is often much higher than that of the OBB task even with the same backbone network [22].

To eliminate the ambiguity of the regression-based OBB, one can treat the OBB-detection problem as a pixel-level classification problem [22]. The OBB is then estimated from the predicted mask by some postprocessing methods. We refer to this process as the segmentation-based OBB representation method. The general segmentation-based OBB representation uses the binary segmentation map, which labels each pixel in the OBB discretely. This binary segmentation map representation is widely used in the field of oriented text detection [23]–[25], which is a single-category oriented object-detection task. We argue that such a binary segmentation map is also unsuitable for many classes such as *harbor, plane, and helicopter* in the aerial images. The pixels of those objects are mainly located around the center of the OBBs. For instance, the OBBs of *harbor* in Fig. 1 contain a number of background pixels, which will affect the convergency of the network. In addition, Jiu *et al.* [26] and Chan-Hon-Tong and Audebert [27] also argued that by removing the ambiguous background pixels, the segmentation network can learn better. To address this ambiguity problem of the background pixels, different from the previous works, we propose a center probability map OBB (CenterMap OBB) to represent the OBB. Unlike the binary segmentation map, the CenterMap OBB represents the probabilities of all pixels being the center of the corresponding OBB. Therefore, the CenterMap OBB is able to reduce the influence of the background pixels in the OBB, and further obtain higher detection performance.

For analyzing the pros and cons of the regression-based OBB and segmentation-based OBB on the oriented object-detection task in the aerial images, we give a comprehensive analysis of different OBB representations in this article. The experimental results on these two kinds of OBB representations demonstrate that the gap of accuracy between the HBB-detection task and the OBB-detection task is greatly reduced by using the segmentation-based OBB, especially CenterMap OBB. In addition, the comparison experiments of Mask OBB and CenterMap OBB on some categories whose pixels are mainly located around the OBBs' center show that the CenterMap OBB can overcome the ambiguity problem of the background pixels. Furthermore, the analysis experiments of false-positives on the five different OBB representations indicate that the CenterMap OBB has the best localization ability.

On the other hand, due to the cluttered background in the aerial images, detectors may detect a large number of uninteresting objects that will inevitably result in increasing false alarms. Therefore, it is important to enhance the features of the interesting objects and suppress the information of the uninteresting objects. Many works use unsupervised attention structures to solve this problem [12], [28], [29], which cannot guide the network to learn the object-level features. In order to capture the features for objects more effectively than the background, we design a weighted pseudosegmentation-guided attention network (WPSGA-Net) to provide object-level features for the detection branch while predicting the HBBs and the OBBs. Specifically, WPSGA-Net uses the semantic-segmentation network to obtain the semantic features of the input image and then uses the RoI Align [30] operation to extract the object-level features from the semantic features. Due to the lack of ground truth for semantic segmentation in the aforementioned data sets, we use OBBs to generate the corresponding ground truth for semantic segmentation and call this ground truth *pseudosegmentation*. To reduce further the influence of the background pixels surrounding the object in the OBB, we apply the CenterMap to generate a weight map for calculating the segmentation loss, which can reduce the contribution of the background pixels.

In this article, we propose the CenterMap-Net by integrating the CenterMap OBB with WPSGA-Net. In particular, the CenterMap-Net can achieve 76.03% and 77.33% mAPs for the OBB task and the HBB task on DOTA, respectively. Meanwhile, it achieves 92.8% and 96.26% mAPs on the HRSC2016 and UCAS-AOD OBB tasks. The contributions of this article are as follows.

1) We propose a CenterMap OBB for better representing the OBB by reducing the influence of background pixels in the OBB. Experiments demonstrate its potential benefit on the DOTA, HRSC2016, and UCAS-AOD data sets.
2) We analyze the pros and cons of the regression-based OBB and segmentation-based OBB representations by lots of extensive experiments.
3) We design a WPSGA-Net to learn the global contexts, aiming at providing object-level features for detection branch to predict HBBs and OBBs more accurately.

The remainder of this article is organized as follows. Section II first describes the related works. The detailed description of the proposed method, including CenterMap OBB and WPSGA-Net, is then provided in Section III. Next, the experimental results are further provided in Section IV. Finally, Section V concludes this article.

II. Related Work

Most of the previous object-detection methods can be roughly divided into two categories: horizontal object-detection and oriented object-detection methods.

A. Horizontal Object-Detection Method

The sliding window paradigm has a long history and dominates the field of object detection in the aerial images. Many

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early detectors are closely related to the handcrafted features such as scale-invariant feature transform (SIFT) [31], Haar-like features [32], histogram of oriented gradients (HOG) [33], local binary patterns (LBP) [34], and deformable part-based models (DPM) [35].

Recently, as the deep convolutional neural network (CNN) is able to learn the robust features of an image, the performance of object detection has been substantially improved whether in the field of general object detection or aerial image object detection. These detectors can be organized into two main categories: two-stage and single-stage detectors.

The two-stage detectors are first reported by the R-CNN [1] that first generates the object proposals by selective search [36] and then classifies and refines the proposal regions by a CNN. For reducing the redundant computation in the R-CNN, the Fast R-CNN [2] extracts the feature of the whole image once and then generates the region features by RoI Pooling [2]. The Faster R-CNN [3] introduces a region proposal network (RPN) to generate efficiently the proposals and allows the detectors to be trained end to end. Based on the Faster R-CNN, the feature pyramid network (FPN) [37] fuses the low-level and high-level features by the top-down pathway and lateral connection to handle the scale problem in object detection. Furthermore, the Mask R-CNN [30] is proposed by adding a mask branch on the Faster R-CNN to obtain pixel-level mask predictions, which also is well known as the instance-segmentation network. Different from the two-stage detectors, single-stage detectors directly predict the class probabilities and bounding box offsets with a single feedforward CNN like YOLO [6], [7], SSD [4], and RetinaNet [8]. They are simpler and more elegant than the two-stage detectors on the network structure and can be optimized end to end directly.

The aforementioned detectors are designed for the field of general object detection. In the field of remote sensing, objects own some distinctive characteristics such as small size, cluttered background, and arbitrary orientation. These problems have been discussed by some recent works. $\mathcal{R}^2$-CNN [12] designs a Tiny-Net backbone and a global attention block to detect tiny objects in the large-scale aerial images. Dong *et al*. [38] propose Sig-NMS to replace traditional nonmaximum suppression (NMS) [39] for improving the precision of detecting small objects. Chen *et al*. [40] propose a multiscale spatial and channelwise attention mechanism to reduce the influence of the cluttered background. However, such horizontal object detectors are still confronted with the challenge of arbitrary orientation in the aerial images. More sophisticated detectors are required for the oriented objects in the aerial images.

B. Oriented Object-Detection Method

Oriented objects are common in aerial images. Deep-learning-based oriented object detectors have been proposed by adopting popular object-detection/segmentation methods like Faster R-CNN [3], SSD [4], RetinaNet [8], and FCN [41]. These oriented object detectors can roughly be divided into two categories: regression-based and segmentation-based methods.

1) Regression-Based Method: The regression-based methods directly regress the OBBs. The work in [42] proposes the rotated region-of-interest pooling layer and the rotated bounding box regression module to detect the oriented ships. Xia *et al*. [15] proposed the FR-O that regresses the offsets of the OBBs relative to the HBBs. ICN [17] joins the image cascade and FPN to extract semantic features for regressing the offsets of the OBBs relative to the HBBs. The RRPN [43] that is designed for text detection presents a rotation RPN to generate prior proposals with the angle information of the object and then regresses the offsets of the OBBs relative to the oriented proposals. R-DFPN [21] adopts the RRPN and puts forward the dense feature pyramid network to solve the narrow width problems of the ships. The RoI-Transformer [18] designs a region-of-interest (RoI) learner to transform the horizontal RoIs to oriented RoIs by a supervised method. All the aforementioned methods rely on an accurate representation of OBB, which we would be further discussed in Section III.

2) Segmentation-Based Method: Segmentation-based methods aim to seek the object region at pixel level and then use the postprocessing methods to obtain the OBBs. Despite its relevance, the segmentation-based oriented object-detection methods in aerial images have been poorly exploited when compared with the field of text detection.

There are some works that adopt the OBB based on the *semantic-segmentation network*. RBMDeepNet [44] proposes a hybrid method to combine the CNN and the restricted Boltzmann machine (RBM) for automatic car extraction in the aerial images. Furthermore, Masouleh and Shahhosseini [44] proposed the Gaussian–Bernoulli RBM for segmenting the vehicles in the aerial images [45]. Audebert *et al*. [46] proposed a segment-before-detect framework to detect vehicles in the aerial RGB images. There are also some other works that adopt the OBB based on the *instance-segmentation network*. Dai *et al*. [47] presented the fused text-segmentation networks to detect and segment the text instance simultaneously. Lyu *et al*. [48] finished the detection and recognition tasks on mask-branch by predicting the word-segmentation map. Mou *et al*. [49] proposed a semantic bounding-aware unified multitask ResFCN to handle vehicle instance segmentation in the aerial images. These segmentation-based methods are restricted to single-category object detection, whereas there are often many different categories to discern in aerial images such as the DOTA data set [15]. To detect multicategory-oriented objects in the aerial images, Wang *et al*. [22] proposed the Mask OBB that uses the binary segmentation map to represent the oriented objects.

Our proposed CenterMap-Net belongs to the segmentation-based method and can handle multicategory object detection in the aerial images. It is based on the instance segmentation framework Mask R-CNN [30].

C. Pseudosegmentation-Guided Attention Network

The attention mechanism has been widely used in vision applications such as image classification [50]–[52], general object detection [53], [54], and pedestrian detection [55]. Inspired by Hu *et al.* [51] and Vaswani *et al.* [56], Li *et al*. [29] proposed a feature-attention FPN that contains

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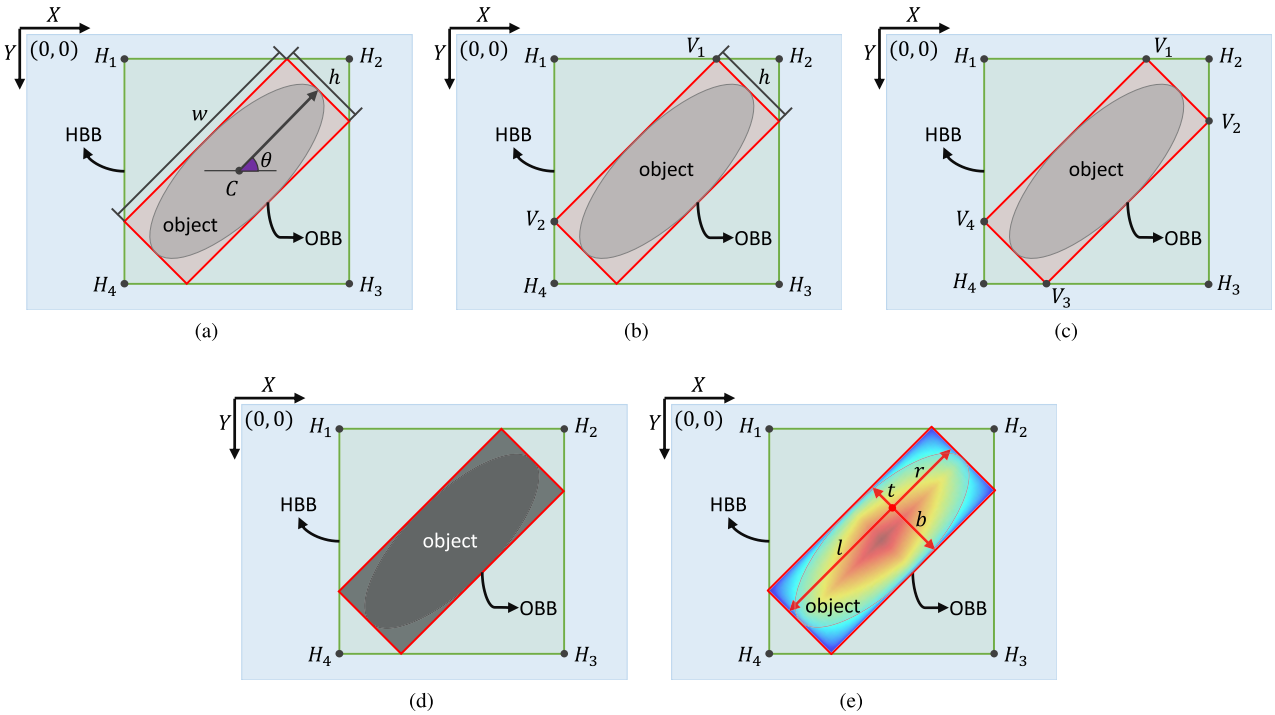


Fig. 2. Five different representations for representing the OBB. (a) θ-based OBB—it can be described by (x, y, w, h, θ), where (x, y), w, h, and θ denote the OBB's center coordinate C, width, height, and angle, respectively. (b) h-based OBB—it can be described by (x_1, y_1, x_2, y_2, h), where (x_1, y_2), (x_2, y_2), and h represent the *first vertex* V_1, *second vertex* V_2, and *height* of the OBB, respectively. (c) Point-based OBB—it can be described by $\{(x_i, y_i)|i = 1, 2, 3, 4\}$, where (x_i, y_i) represents the ith vertex of the OBB. (d) Mask OBB—it represents the OBB by a binary segmentation map, where the values for the whole OBB pixels are **1**. (e) CenterMap OBB—it represents the OBB by a CenterMap, where the values for the border/center pixels are **0/1** and the values of other pixels gradually decay from center to border. Note that (a)–(c) are the regression-based OBB representations and (d) and (e) are the segmentation-based OBB representations.

channelwise attention and pixelwise attention to effectively capture the foreground information and restrain the background in the aerial images. CAD-Net [28] designs a spatial-and-scale-aware attention module to guide the network to focus on the regions that are more informative. The work in [57] that is inspired by the convolutional block attention module [52] proposes a dense attention pyramid network to highlight the salient object features for improving the ship-detection accuracy in the synthetic aperture radar images. The work in [40] proposes a multiscale spatial and channelwise attention mechanism to make the network pay more attention to the object in the aerial images as human vision. SDS-RCNN [58] proposes a segmentation-infusion network to enable joint supervision on semantic segmentation and detection. SCRDet [59] proposes a multidimensional attention network that uses the supervised pixel and the channel-attention network to suppress the background noise and highlight the object features. Both SDS-RCNN and SCRDet add a supervised semantic-segmentation branch to infuse the image-level features into the backbone network layers.

Different from the aforementioned methods, our proposed WPSGA-Net uses a weight map that is generated by CenterMap to restrain the background noise. At the same time, WPSGA-Net provides object-level features for the detection branch while predicting the HBBs and OBBs. More details about WPSGA-Net will be introduced in Section III.

III. Methodology

In this section, the details of the proposed method will be introduced. At first, by analyzing different OBB representations, the new representation CenterMap OBB is proposed. Then, the overall architecture of the proposed network is derived. Next, the WPSGA-Net is elaborated. Finally, the loss function is given.

A. Representations for Oriented Bounding Box

As mentioned in Section II, the representations for OBB can be divided into two categories: regression-based representation and segmentation-based representation. Fig. 2 demonstrates these representations. Specifically, Fig. 2(a)–(c) shows the regression-based representations and Fig. 2(d) and (e) shows the segmentation-based representations. For analyzing the pros and cons of these OBB representations, we reimplement these OBB representations on the same network structure. The details of these representations and implementations are described as follows.

1) θ-Based OBB Representation: θ-based OBB [18], [43] can be described by a 5-D vector (x, y, w, h, θ), where (x, y), w, h, and θ denote the OBB's center coordinate, width, height, and angle. θ $(0° \leq \theta < 180°)$ is defined as its length-wise direction and horizontal direction. This representation is depicted in Fig. 2(a). To encode a ground-truth θ-based OBB with respect to its matching proposal generated by the RPN,

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the following encoding functions are used:

$$\begin{aligned}\phi_x &= 10 \cdot (x^* - x^{\mathrm{p}})/w^{\mathrm{p}}, \quad \phi_y = 10 \cdot (y^* - y^{\mathrm{p}})/h^{\mathrm{p}} \\ \phi_w &= 5 \cdot \log(w^*/w^{\mathrm{p}}), \quad \phi_h = 5 \cdot \log(h^*/h^{\mathrm{p}}) \\ \phi_\theta &= 5 \cdot \log(\theta^*/\theta^{\mathrm{p}}) \end{aligned} \tag{1}$$

where x^{p} and x^* are the horizontal proposals generated by the RPN and ground truth, respectively (likewise for y, w, h, and θ). Note that the scalars 10 and 5 are typically used in previous works [60], where the scalar 10 means the standard deviations of x and y are 0.1 and the scalar 5 means the standard deviations of w, h, and θ are 0.2, the same hereinafter.

2) h-Based OBB Representation: h-based OBB [61] can also be described by a 5-D vector (x_1, y_1, x_2, y_2, h), namely, first vertex $V_1 = (x_1, y_1)$, second vertex $V_2 = (x_2, y_2)$, and height h of the OBB. This representation is depicted in Fig. 2(b). To encode a ground-truth h-based OBB with respect to its matching proposal generated by the RPN, the following encoding functions are used:

$$\begin{aligned}\phi_{x_1} &= 10 \cdot \left(x_1^* - x_1^{\mathrm{p}}\right)/w^{\mathrm{p}}, \quad \phi_{y_1} = 10 \cdot \left(y_1^* - y_1^{\mathrm{p}}\right)/h^{\mathrm{p}} \\ \phi_{x_2} &= 10 \cdot \left(x_2^* - x_2^{\mathrm{p}}\right)/w^{\mathrm{p}}, \quad \phi_{y_2} = 10 \cdot \left(y_2^* - y_2^{\mathrm{p}}\right)/h^{\mathrm{p}} \\ \phi_h &= 5 \cdot \log(h^*/h^{\mathrm{p}}) \end{aligned} \tag{2}$$

where x_1^{p} and x_1^* are the horizontal proposals generated by the RPN and the ground truth, respectively (likewise for y_1, x_2, y_2, h).

3) Point-Based OBB Representation: The point-based OBB [15], [17], [19] can be described by an 8-D vector $(x_1, y_1, x_2, y_2, x_3, y_3, x_4, y_4)$, namely, four vertices $\{V_i = (x_i, y_i) | i = 1, 2, 3, 4\}$ of OBB. This representation is depicted in Fig. 2(c). To encode a ground-truth point-based OBB with respect to its matching proposal generated by the RPN, the following encoding functions are used:

$$\begin{aligned}\phi_{x_1} &= 10 \cdot \left(x_1^* - x_1^{\mathrm{p}}\right)/w^{\mathrm{p}}, \quad \phi_{y_1} = 10 \cdot \left(y_1^* - y_1^{\mathrm{p}}\right)/h^{\mathrm{p}} \\ \phi_{x_2} &= 10 \cdot \left(x_2^* - x_2^{\mathrm{p}}\right)/w^{\mathrm{p}}, \quad \phi_{y_2} = 10 \cdot \left(y_2^* - y_2^{\mathrm{p}}\right)/h^{\mathrm{p}} \\ \phi_{x_3} &= 10 \cdot \left(x_3^* - x_3^{\mathrm{p}}\right)/w^{\mathrm{p}}, \quad \phi_{y_3} = 10 \cdot \left(y_3^* - y_3^{\mathrm{p}}\right)/h^{\mathrm{p}} \\ \phi_{x_4} &= 10 \cdot \left(x_4^* - x_4^{\mathrm{p}}\right)/w^{\mathrm{p}}, \quad \phi_{y_4} = 10 \cdot \left(y_4^* - y_4^{\mathrm{p}}\right)/h^{\mathrm{p}} \end{aligned} \tag{3}$$

where x_1^{p} and x_1^* are the horizontal proposals generated by the RPN and the ground truth, respectively (likewise for $y_1, x_2, y_2, x_3, y_3, x_4, y_4$).

4) Mask OBB Representation: Although the aforementioned regression-based OBB representations can ensure the uniqueness in normal conditions with some rules, they still allow some extreme conditions. In these conditions, a tiny change of the OBB angle would result in a large change of OBB representation. We denote the angle value in these conditions as the discontinuity point. For the oriented object detectors, similar features extracted by the detector with close positions are supposed to generate similar position representations. However, the OBB representations of these similar features would differ greatly near the discontinuity points. This would force the detector to learn totally different position representations for similar features. It would impede the detector training process and deteriorate the detector's performance obviously.

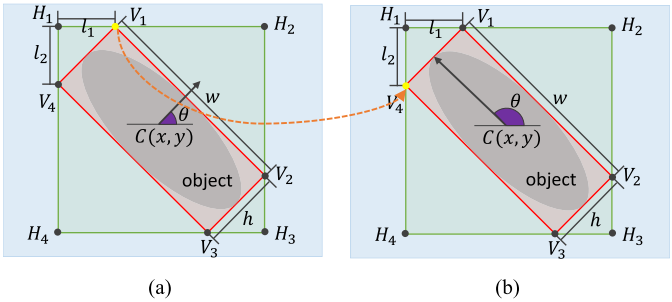


Fig. 3. The ambiguity condition of two similar OBBs. In both (a) and (b), $H_i = (x_i, y_i) | i = 1, 2, 3, 4$ are the four vertices of the HBB, $V_i = (x_i, y_i) | i = 1, 2, 3, 4$ are the four vertices of the OBB, and $C = (x, y)$ is the center coordinate of the OBB. (a) When $l_1 < l_2$, the yellow vertex will be the "Top Left" vertex of the OBB by the definition in [15]. By this definition, the point-based OBB, h-based OBB, and θ-based OBB can be described as (V_1, V_2, V_3, V_4), (x_1, y_1, x_2, y_2, h), and (x, y, w, h, θ). (b) When $l_1 > l_2$, the "top left" vertex of the OBB will be transformed from V_1 to V_4. Because this transformation, the representations of the regression-based OBB will change greatly. In this condition, the point-based OBB, h-based OBB, and θ-based OBB need to be described as (V_4, V_1, V_2, V_3), (x_4, y_4, x_1, y_1, w), and $(x, y, h, w, \theta + \pi/2)$.

Fig. 3 shows one ambiguity condition of the regression-based OBB representations intuitively, while the angle of object relative to HBB is near 45° or 135°. Specifically, for the point-based OBB representation, to ensure the uniqueness of OBB definition, Xia *et al.* [15] chose the vertex closest to the "top left" vertex of the corresponding HBB as the first vertex. Then, the other vertices are fixed in the clockwise order, so we get the unique representation of the OBB. Nevertheless, this representation allows discontinuity point, as illustrated in Fig. 3. When $l_1 < l_2$, the OBB is represented with $R_1 = (V_1, V_2, V_3, V_4)$, as shown in Fig. 3(a). Otherwise, the OBB is represented with $R_2 = (V_4, V_1, V_2, V_3)$, as shown in Fig. 3(b). When l_2 increases with θ, until θ approaches and surpasses $\pi/4$, the OBB representation would jump from R_1 to R_2, and vice versa. Hence, $\pi/4$ is a discontinuity point in this mode. For the h-based OBB and the θ-based OBB, $\pi/4$ is still a discontinuity point for both of them. With θ oscillating near $\pi/4$, the h-based OBB representation would switch between (x_1, y_1, x_2, y_2, h) and (x_4, y_4, x_1, y_1, w). The θ-based OBB representation would switch back and forth between (x, y, w, h, θ) and $(x, y, h, w, \theta + \pi/2)$ similarly.

For handling the ambiguity problem in the regression-based OBB representations, Wang *et al.* [22] treat the OBB-detection problem as the pixel-level classification problem that is called segmentation-based OBB representation. However, the aerial image-detection data sets like DOTA [15] and UCAS-AOD [13] give the point-based OBB as the ground truth, while HRSC2016 [14] gives the θ-based OBB as the ground truth. However, for the pixel-level classification, pixel-level annotations are essential. In order to handle this problem, pixel-level annotations are converted from original OBB ground truth. Specifically, the pixels inside the OBB are labeled as positive and the pixels outside are labeled as negative. Wang *et al.* [22] called this OBB representation as Mask OBB. Fig. 2 illustrates the original regression-based OBB and the converted Mask OBB. In Fig. 2(d), the black region inside the point-based OBB

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TABLE I
NUMBER OF OBJECTS PER SELECTED CATEGORY ON DOTA TRAINVAL SET

PL	BD	SH	ST	RA	HA	HC	Total	Rate
10,586	629	37,028	7,917	578	8,073	703	65,514	51.25%

is Mask OBB. Different from the regression-based OBBs, the Mask OBB is unique in the definition. Using the Mask OBB representation, the problem of ground-truth ambiguity can be solved in nature and there is no discontinuity point allowed in this mode.

In the inference stage, the predicted Mask OBB will be converted into the corresponding regression-based OBB according to the data set by the topological structural analysis algorithm [62].

5) CenterMap OBB Representation: As shown in Fig. 2(d), there are still lots of background pixels surrounding *real object* in the Mask OBB when the *real object* is not a strict quadrilateral, such as *plane, harbor*, and *helicopter* in the aerial images. In these objects, foreground pixels are concentrated on the center of the OBB, while background pixels are concentrated on the boundary of the OBB. However, these background pixels might affect the convergency of the segmentation network, resulting in the low-quality segmentation results. We call this as the ambiguity problem of the background pixels. In order to show the necessity of solving this problem, we count the number of categories that might contain background pixels in the OBBs on the DOTA trainval set, including: *plane (PL), baseball diamond (BD), ship (SH), storage tank (ST), roundabout (RA), harbor (HA), and helicopter (HC)*. The statistical results are shown in Table I, about 51.25% of objects might contain background pixels on the DOTA trainval set. Thus, it is necessary to weaken the background pixels and enhance the center pixels of the objects in the aerial images.

To handle the ambiguity problem of the background pixels, we propose a CenterMap OBB for better representing the oriented objects in the aerial images. Unlike Mask OBB that labels each pixel discretely, the CenterMap OBB encodes OBB with a CenterMap, where the values for the border/center pixels are **0/1** and the values of other pixels gradually decay from center to border. As shown in Fig. 2(e), the redder color means greater value and the bluer color means smaller value. Since the CenterMap OBB representation is also based on the segmentation paradigm, the problem of target definition ambiguity in the regression-based OBB representations can also be handled by the CenterMap OBB representation.

There are many methods to generate the CenterMap OBB such as Gaussian distribution. In order to make the values at the border of CenterMap OBB exactly equal to **0** and the value at the center of CenterMap OBB is exactly equal to **1**, inspired by Tian *et al.* [63], we define the center probability $p_{x,y}$ of pixel at (x, y) as

$$p_{x,y} = \left(\frac{\min(l,r)}{\max(l,r)} \times \frac{\min(t,b)}{\max(t,b)}\right)^{1/k} \tag{4}$$

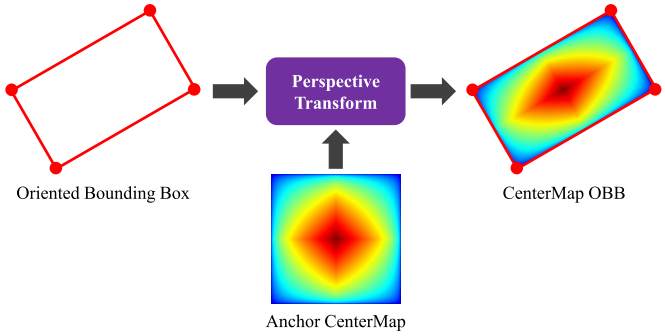


Fig. 4. Illustration of CenterMap OBB generation procedure. The Anchor CenterMap is a CenterMap with a fixed size and shape. Perspective transform wraps the Anchor CenterMap to the shape of OBB.

where l, t, r, and b are the distances from the location (x, y) to the four boundaries (left, top, right, and bottom) of the OBB, as shown in Fig. 2(e). k is a key hyperparameter to adjust the shape of CenterMap OBB, which is called *shape factor* and will be discussed later. If a pixel falls into multiple OBBs, we choose the OBB with a minimal area as its probability for keeping small objects.

However, it is inefficient when (4) is directly used to generate CenterMap OBB for each OBB. To address this problem, we define an Anchor CenterMap as a template and use the perspective transform to wrap Anchor CenterMap to the OBB. Fig. 4 shows the generation pipeline of CenterMap OBB for a point-based OBB. The generated CenterMap OBB will be the ground truth for the OBB-detection network.

B. Overall Architecture

The overall architecture (see Fig. 5) of our method is based on the Mask R-CNN [30]. We modify the mask branch in the Mask R-CNN to the oriented bounding-detection branch (OBB branch). Specifically, the input image is first fed into a backbone network to extract deep features and generate region proposals by the RPN. Then, the regional features extracted by RoI Align [30] on the proposals are fed into the HBB branch and the OBB branch to generate the HBBs and OBBs.

1) Backbone: We apply the FPN [37] with ResNet [64] as the backbone and call the final feature map set of the FPN as $\{P_2, P_3, P_4, P_5, P_6\}$. Note that $\{P_2, P_3, P_4, P_5, P_6\}$ have the receptive fields of $\{4, 8, 16, 32, 64\}$ pixels with respect to the input image.

2) Region Proposal Network: The RPN [3] is used to generate region proposals on each level of FPN. Following the original FPN [37], we assign the anchors of a single scale to each level. Specifically, we set five anchors with the areas of $\{32^2, 64^2, 128^2, 256^2, 512^2\}$ pixels on five levels $\{P_2, P_3, P_4, P_5, P_6\}$, respectively. Different anchor aspect ratios $\{1:2, 1:1, 2:1\}$ are also adopted at each level. Thus, there are a total of 15 anchors over the pyramid.

3) Head: RoI Align [30] is adapted to extract the regional features of the proposals produced by the RPN, since it can preserve accurate location information, which is quite beneficial to the segmentation task in the OBB branch. Through RoI Align, all regional features are resized to 7×7 for the

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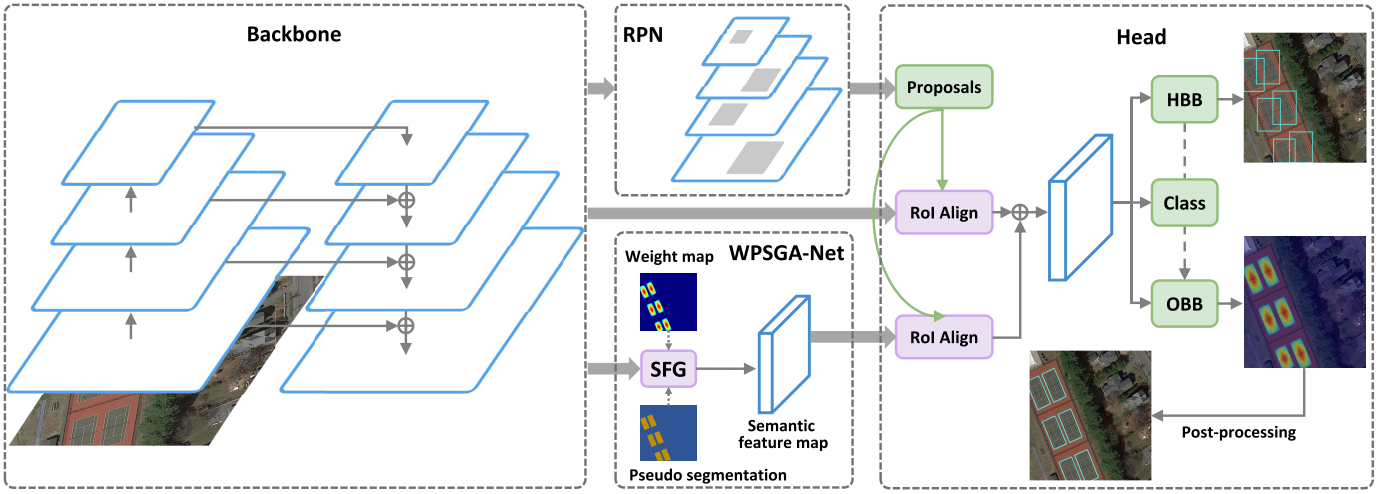


Fig. 5. Framework of the proposed CenterMap-Net. Backbone is the ResNet with FPN, RPN is the RPN, WPSGA-Net is the weighted pseudosegmentation guided attention network, SFG is the semantic feature generation module, and HBBs and OBBs are generated by the HBB branch and the OBB branch in the head, respectively.

HBB branch and 14×14 for the OBB branch. Then, the 7×7 features are used to regress the HBB and classify the objects in HBB branch, while the 14×14 features are used to predict 28×28 masks in the OBB branch.

C. Weighted Pseudosegmentation-Guided Attention Network

To help further the detection network to extract the interesting objects from the cluttered background in the aerial images, we design a WPSGA-Net to extract the object-level features from the whole image. The feature-extraction module, which is also called semantic feature generation (SFG) module, is the key in WPSGA-Net. Inspired by Kirillov *et al.* [65] and Chen *et al.* [66], we use the encoder–decoder architecture to design the SFG. Specifically, higher level feature maps with context information as well as lower level feature maps with location information, as the outputs of the FPN, are incorporated to generate the semantic features. Fig. 6 illustrates the details. Starting from the highest FPN level P_5, we perform an upsampling operation to yield a feature map, where the upsampling operation consists of 3×3 convolution, group normalization [67], ReLU [68], and $2\times$ bilinear upsampling. Here, we give a detailed formulation for this upsampling operation

$$f_u = \mathrm{BU}\{\sigma(\mathrm{GN}(f(F^i, w)))\} \tag{5}$$

which means the upsampling operation (conv $\rightarrow 2\times$ in Fig. 6), F^i denotes the input feature, $w \in \mathbb{R}^{C_F \times C_F \times 3 \times 3}$ indicates the convolutional parameters, $f(.,.)$ denotes a 3×3 convolution operation, GN denotes the group normalization, σ represents the ReLU-activation function, and BU denotes the $2\times$ bilinear upsampling operation. For the input feature map $P_i \in \mathbb{R}^{C_i \times W_i \times H_i}, i = 3, 4, 5$ from the ith level of FPN, the upsampled feature map M_i can be calculated as

$$M_i = \underbrace{f_u(\cdots, f_u(P_i))}_{i-2}. \tag{6}$$

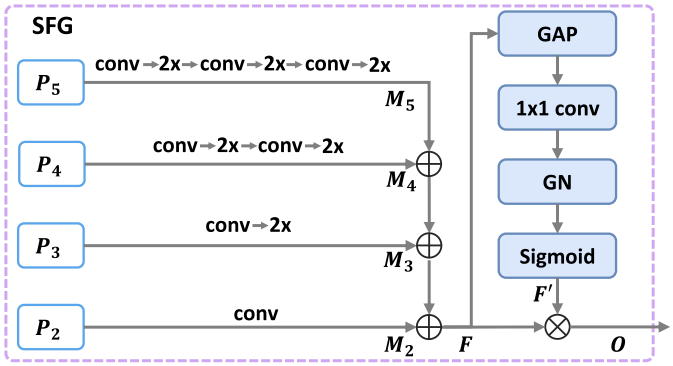


Fig. 6. Illustration of the SFG module. SFG upsamples the outputs of the FPN to obtain the semantic-segmentation features. P_2, P_3, P_4, P_5 are the outputs of the FPN, conv $\rightarrow 2\times$ is the upsampling operation, M_2, M_3, M_4, M_5 are the upsampled feature maps, GAP and GN indicate global average pooling and group normalization, respectively, and $\oplus$ and $\otimes$ denote the spatial elementwise sum and multiplication, respectively.

Note that we keep the spatial scale of P_2, so there is no $2\times$ upsampling operation on the P_2. The results are a set of feature maps at the same scale, which are then elementwise-summed. A final $2\times$ bilinear upsampling operation is used to generate the feature map $F \in \mathbb{R}^{C_s \times W_s \times H_s}$.

Motivated by the squeeze-and-excitation networks [51], we add a channel-attention module to downweight useless the background channels. The reweighting operator $F' \in \mathbb{R}^{C_s \times 1 \times 1}$ can be generated as

$$F' = \mathrm{Sigmoid}\{\mathrm{GN}(f(\mathrm{G}(F^i, w)))\} \tag{7}$$

$$O_m = F'_m \otimes F_m \tag{8}$$

where Sigmoid denotes the sigmoid function, G and GN, respectively, denote the global average pooling and group normalization, $w \in \mathbb{R}^{C_s \times C_s \times 1 \times 1}$ represents the convolutional parameter, m indicates the mth pixel channel in the corresponding feature map, $\otimes$ denotes the spatial elementwise multiplication, O_m is the final feature map of SFG

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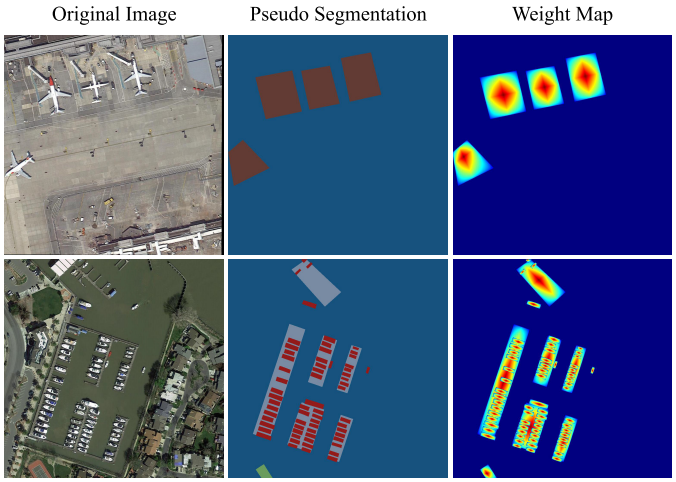


Fig. 7. Pseudosegmentation and weight map. First column: original images that come from the DOTA data set. Second column: corresponding pseudosegmentations—different colors mean different categories. Third column: corresponding weight maps. Best viewed in color.

for the mth channel, and O is the final feature map of SFG. Then, one 1×1 convolution is used on the feature map O to generate final *label map* L and *semantic feature map* S.

The L is used to calculate the loss of WPSGA-Net. The ground truth for WPSGA-Net is generated by the OBB ground truth, since the aerial image-detection data sets have no semantic-segmentation ground truth. As such, we call the generated ground truth as *pseudosegmentation*. Specifically, in the whole image, the pixels inside the OBB are labeled as certain class of the OBB and the pixels outside the OBB are labeled as background. Fig. 7 illustrates the original images and their corresponding pseudosegmentation.

In addition, the S is fused with the HBB branch feature map and the OBB branch feature map separately. Specifically, given a proposal from the RPN, we apply RoI Align to extract a semantic feature patch and a proposal feature patch of the same shape (e.g., 7×7 for HBB branch and 14×14 for OBB branch) from the S and the corresponding level of the FPN, respectively. The semantic feature patch and the proposal feature patch are then combined by the elementwise sum to generate new region features for the HBB branch and the OBB branch.

However, as mentioned in Section I, many background pixels are treated as positive class that will affect the convergency of WPSGA-Net. For handling this problem, we design a weight map by CenterMap to downweight the contributions of these pixels to the total loss that will be described in detail in Section III-D. Fig. 7 demonstrates the original images and their corresponding weight maps.

D. Loss Function

The whole network is optimized by a joint loss function $\mathcal{L}$ during the training stage

$$\mathcal{L} = \mathcal{L}_{\text{RPN}} + \mathcal{L}_{\text{Head}} + \mathcal{L}_{\text{Seg}} \tag{9}$$

where $\mathcal{L}_{\text{RPN}}$ is the RPN loss [3] and $\mathcal{L}_{\text{Head}}$ is the head loss of the network

$$\begin{aligned}\mathcal{L}_{\text{Head}} = {} & \alpha_1 \frac{1}{N_{\text{cls}}} \sum_i \mathcal{L}_{\text{cls}}\left(c_i, c_i^*\right) \\ & + \alpha_2 \frac{1}{N_{\text{reg}}} \sum_i \left[c_i^* \geq 1\right] \mathcal{L}_{\text{hbb}}\left(\text{HBB}_i, \text{HBB}_i^*\right) \\ & + \alpha_3 \frac{1}{N_{\text{reg}}} \sum_i \left[c_i^* \geq 1\right] \mathcal{L}_{\text{obb}}\left(\text{OBB}_i, \text{OBB}_i^*\right)\end{aligned} \tag{10}$$

where $\mathcal{L}_{\text{cls}}$ is the cross-entropy loss, c_i and c_i^* denote the probability distributions of the predicted classes and the ground-truth classes. $\mathcal{L}_{\text{hbb}}$ is smooth L1 loss [2], where HBB_i and HBB_i^* denote the predicted and targeted regression offsets of the HBB, respectively. $\mathcal{L}_{\text{obb}}$ is related to the OBB representation method. In the CenterMap-Net, we use the mean-squared-error (MSE) loss as the loss function of the OBB branch

$$\mathcal{L}_{\text{obb}} = \sum_{m=1}^{28} \sum_{n=1}^{28} \left\| \text{OBB}_{m,n} - \text{OBB}_{m,n}^* \right\|_2^2 \tag{11}$$

where $\text{OBB}_{m,n}$ and $\text{OBB}_{m,n}^*$, respectively, denote the predicted CenterMap and target CenterMap at pixel position (m, n). $\mathcal{L}_{\text{Seg}}$ is the loss of WPSGA-Net. As mentioned in Section III-C, for reducing the influence of the background pixels around the positives, WPSGA-Net applies a weight map to downweight the contribution of these pixels. Thus, the segmentation loss can be formulated as

$$\mathcal{L}_{\text{Seg}} = \alpha_4 \sum_{i=1}^{W/2} \sum_{j=1}^{H/2} w_{ij} (1 - \hat{p}_{ij})^{\gamma} \log(\hat{p}_{ij}) \tag{12}$$

where

$$\hat{p}_{ij} = \begin{cases} p_{ij}, & \text{otherwise} \\ 1 - p_{ij}, & \text{if } y_{ij} = 0 \end{cases} \tag{13}$$

$$w_{ij} = \begin{cases} 1 + \left(\dfrac{\min(l, r)}{\max(l, r)} \times \dfrac{\min(t, b)}{\max(t, b)} \right)^{1/k}, & \text{otherwise} \\ 1, & \text{if } y_{ij} = 0 \end{cases} \tag{14}$$

where $p_{ij} \in [0, 1]$ is the estimated probability for classes in location (i, j), $y_{ij} \in \{0, 1, \ldots, N\}$, N is the number of classes, and $y_{ij} = 0$ represents the background. γ is the hyperparameter in focal loss [8], and we experimentally set $\gamma = 2$, as suggested in [8]. w_{ij} is the weight map generated by CenterMap to reduce the background pixel contributions and k is experimentally set to 4. $\alpha_1, \alpha_2, \alpha_3$, and α_4 are the weights for classification, HBB regression, OBB regression, and WPSGA-Net losses.

The ground truth of the OBB branch is described in Section III-A. We calculate OBB's HBB as the ground truth of the RPN and HBB branches. The ground truth of WPSGA-Net is described in Section III-C.

IV. Experiments

In this section, we first introduce the data sets, evaluation metric, and implementation details, then analyze the pros and

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cons of different OBB representations, and finally demonstrate the performance of our proposed method together with some analysis.

A. Data Sets and Implementation Details

Our proposed method is evaluated over three oriented object-detection data sets in the remote sensing field, including the DOTA [15] data set, HRSC2016 [14] data set, and UCAS-AOD [13] data set.

1) DOTA Data Set: DOTA is a data set for multiclass object detection in the aerial images. It contains 2806 images with scales from 800 × 800 to 4000 × 4000 pixels. There are 15 object categories: *baseball diamond (BD), bridge (BR), basketball court (BC), ground track field (GTF), helicopter (HC), harbor (HA), large vehicle (LV), plane (PL), roundabout (RA), small vehicle (SV), storage tank (ST), soccer ball field (SBF), swimming pool (SP), ship (SH), and tennis court (TC)*. Each object in this data set is annotated with the point-based OBB. The training, validation, and test sets include 1/2, 1/6, and 1/3 of the data set, respectively. DOTA aims for two tasks: HBB task and OBB task.

2) HRSC2016 Data Set: HRSC2016 is a data set for ship detection in the aerial images. There are 1061 images that vary from about 300 × 300 to 1500 × 900 pixels in HRSC2016. The training, validation, and test sets contain 436, 181, and 444 images, respectively. Objects are annotated with the θ-based OBB.

3) UCAS-AOD Data Set: UCAS-AOD is a data set for plane and car detection in the aerial images. It contains 1510 images with 14 595 objects annotated with the point-based OBB. The scale of all images is 1280 × 659 pixels. The data set is randomly split into 1110 training and 400 testing images.

4) Evaluation Metric: We adopt the mean average precision (mAP) as the evaluation metric in all experiments, since mAP has been widely used for the performance evaluation of multicategory object detection. For the DOTA data set, we submit our results on the test set to the official evaluation server to obtain the mAPs. For HRSC2016 and UCAS-AOD datasets, we report the standard VOC12 AP metric with IoU threshold of 0.5.

5) Data Preprocessing: Images in the DOTA data set have huge image sizes (e.g., 4000 × 4000 pixels). For fitting the limited hardware memory, we crop the patches of 1024 × 1024 pixels from the original images with an overlap of 200 pixels in both training and inference stages. Furthermore, we do not use the data-augmentation and image pyramid scheme technologies except randomly flipped along horizontal direction with a probability of 0.5, unless specified otherwise.

6) Implementation Details: All models are implemented in PyTorch. We use ResNet-50 [64], which is pretrained on the ImageNet [69] with FPN [37] as the backbone, unless specified otherwise. By default, models are trained in a batch size of 8 on 4 NVIDIA Titan Xp GPUs. Models are trained in 12 epochs by default, with the learning rate starting from 0.01 and decreased by a factor of 0.1 at epochs 8 and 11. Stochastic gradient descent with the weight decay of 0.0001 and the momentum of 0.9 is used in all experiments. In addition, the batch sizes of RPN and Fast R-CNN are set to 256 and 512 per image, with a sample ratio 1 : 3 of positive to negatives. The proposal number of the RPN is set to 2000, and we run the HBB branch on these proposals, following NMS. The OBB branch is then applied to 500 HBBs with the highest scores.

TABLE II
MAPS OF DIFFERENT OBB BRANCH LOSS WEIGHTS α_3. ($\alpha_1 = 1, \alpha_2 = 1, \alpha_4 = 1, and k = 4$)

α_3	**0.1**	**0.3**	**1**	**3**	**10**
OBB (%)	66.56	67.38	68.81	70.99	69.88
HBB (%)	70.39	70.48	70.89	70.94	70.06

TABLE III
MAPS OF DIFFERENT VALUES OF CENTERMAP *Shape Factor k*. ($\alpha_1 = 1, \alpha_2 = 1, \alpha_3 = 3, and \alpha_4 = 1$)

k	**2**	**4**	**8**	**16**	**32**
OBB (%)	67.96	70.99	70.76	70.48	70.04
HBB (%)	70.77	70.94	70.58	70.77	70.13

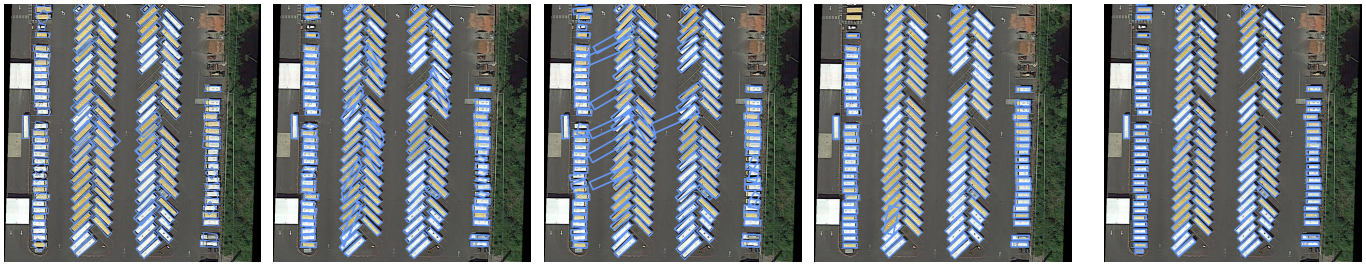
The loss weight α_3 and *shape factor k* are two key hyperparameters to adjust the contribution of the OBB branch loss and the CenterMap OBB shape. We run a couple of experiments to choose the appropriate values of α_3 and k. In our experiments, we set *shape factor* $k = 4$ and select α_3 from 0.1 to 10. As shown in Table II, we can see that 3 is a turning point of accuracy. Thus, we empirically set α_3 as 3. Then, we remain α_3 unchanged and change the value of k from 2 to 32. As shown in Table III, we find that $\alpha_3 = 3$ and $k = 4$ are an optimal parameter combination. The loss weights of α_1 and α_2 are both set to 1, as suggested in [3]. In addition, the loss weight α_4 of WPSGA-Net is simply set to 1, since it does not affect the overall performance in a large range in our experiments.

B. Comparison of Different OBB Representations

1) Analysis of Accuracy: OBBs can be represented in a variety of ways, as shown in Fig. 2. In this article, we reimplement the aforementioned five OBB representations on the same basic network structure for a fair comparison. Different from the CenterMap OBB representation that uses MSE loss as the OBB branch loss function, θ-based OBB, h-based OBB, and point-based OBB representations use smooth L1 loss as the loss function and the Mask OBB representation uses the cross-entropy loss function. Note that the OBB branch cannot directly generate final OBBs for accuracy evaluation while using segmentation-based OBB representations. Thus, the topological structural analysis algorithm [62] of finding the oriented rectangle with the minimum area enclosing the mask area is applied.

As shown in Table IV, we find that there is a higher gap between the HBB and OBB performances for the regression-based OBB representations than the segmentation-based OBB representations. Theoretically, changing from the prediction of HBB to OBB should not affect the classification precision. We argue that the reduction is due to the low-quality localization, which is caused by

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(a) (b) (c) (d) (e)

Fig. 8. (a)–(c) Results from the regression-based OBB representations (θ-based OBB, point-based OBB, and h-based OBB) on the DOTA data set. (d) and (e) Results from the segmentation-based OBB (Mask OBB and CenterMap OBB). Best viewed in zoomed-in view.

TABLE IV

COMPARISON WITH DIFFERENT OBB REPRESENTATION METHODS ON THE GAP OF MAPS BETWEEN THE HBB AND THE OBB OVER THE DOTA TEST SET (%). THE BEST RESULT IS HIGHLIGHTED IN BOLD

OBB Representation	Backbone	OBB	HBB	Gap
θ-based OBB	ResNet-50-FPN	69.40	70.22	0.82
point-based OBB	ResNet-50-FPN	69.43	70.65	1.22
h-based OBB	ResNet-50-FPN	67.45	70.46	3.01
Mask OBB	ResNet-50-FPN	69.97	70.14	0.17
CenterMap OBB	ResNet-50-FPN	**70.99**	**70.94**	**0.05**

TABLE V

COMPARISON OF DIFFERENT OBB REPRESENTATION METHODS ON TRAINING SPEED, INFERENCE SPEED, POSTPROCESSING SPEED, AND MEMORY

OBB Representation	Training (s/iter)	Inference (FPS)	Post-processing (FPS)	Param (MB)
θ-based OBB	0.43	8.7	91.6	442.7
point-based OBB	0.43	8.9	89.3	443.1
h-based OBB	0.43	9.0	83.0	442.7
Mask OBB	0.58	6.2	7.6	351.8
CenterMap OBB	0.58	6.3	7.6	351.9

the discontinuity point, as discussed in Section III-A. There should not be such a large gap between the performance of the HBB and OBB tasks if the representation of OBB is defined well. The results of the regression-based OBB representation verified that. In addition, the CenterMap OBB achieves higher mAP on both OBB and HBB tasks and a lower gap between the OBB task and the HBB task than the Mask OBB. More analysis between the CenterMap OBB and the Mask OBB is given in Section IV-B3.

Fig. 8 shows the visualization of the detection results using different OBB representations. We can see that regression-based OBB representations tend to produce many incorrect detection results. As a comparison, the Mask OBB and CenterMap OBB representations are capable of correctly detecting those objects, and the CenterMap OBB can detect more objects than the Mask OBB, as shown in Fig. 8(d) and (e) (top left).

2) Analysis of Computing Time: For timing the models, we use a machine with 128-GB RAM, Intel Xeon E5-2620 v4 processor, and one NVIDIA Titan Xp GPU card. The images used for timing are DOTA images with the image scale 1024×1024 pixels. We average the timings over 10 833 images. The training speed is measured with *second/iteration (s/iter)*, which is the time of updating the network parameters once. The inference speed is measured with frames per second (FPS), which excludes postprocessing. For the regression-based OBB representations, postprocessing time mainly includes NMS, which runs on the GPU. For the segmentation-based OBB representations, postprocessing time includes NMS and finding the oriented rectangle on the predicted mask. As can be seen in Table V, segmentation-based OBB representations (Mask OBB and CenterMap OBB) use less memory, since regression-based OBB representations use a fully connected layer in the OBB branch. In the training stage, segmentation-based OBB representations need more time to update the network parameters, since they need more time to compute the loss of the OBB branch. In the inference stage, because regression-based OBB representations do not need to encode the predicted mask to the run-length encoding (RLE) format, they are faster than the segmentation-based OBB representations that need to convert the predicted mask into the enclosing oriented rectangles. To sum up, the models can obtain greater performance but need to take more processing time by using the segmentation-based OBB representations.

3) Analysis of Background Pixels: As mentioned in Section III-A5, the CenterMap OBB is proposed to handle the ambiguous background pixels in the OBBs. To analyze the influence of the background pixels in the OBBs, we show the comparison between the Mask OBB and the CenterMap OBB on the AP metric for each category in Table VI. The selected categories include *plane, baseball diamond, ship, storage tank, roundabout, harbor, and helicopter* that are not the strict quadrilaterals and cannot be well described by the OBBs. Observing that the CenterMap OBB all exceeds the Mask OBB with the same backbone network in these categories, which indicates the results well fit with our motivations, and we can find that the CenterMap OBB is better than the Mask OBB for describing the oriented objects in the aerial images.

4) Analysis of False-Positives: In order to reveal the failure reasons of the five different OBB representations, we conduct experiments on the distributions and trends of the top-ranked false-positive (FP) types in the DOTA validation set, as shown in Fig. 9(a)–(e). Note that the corresponding APs are shown in

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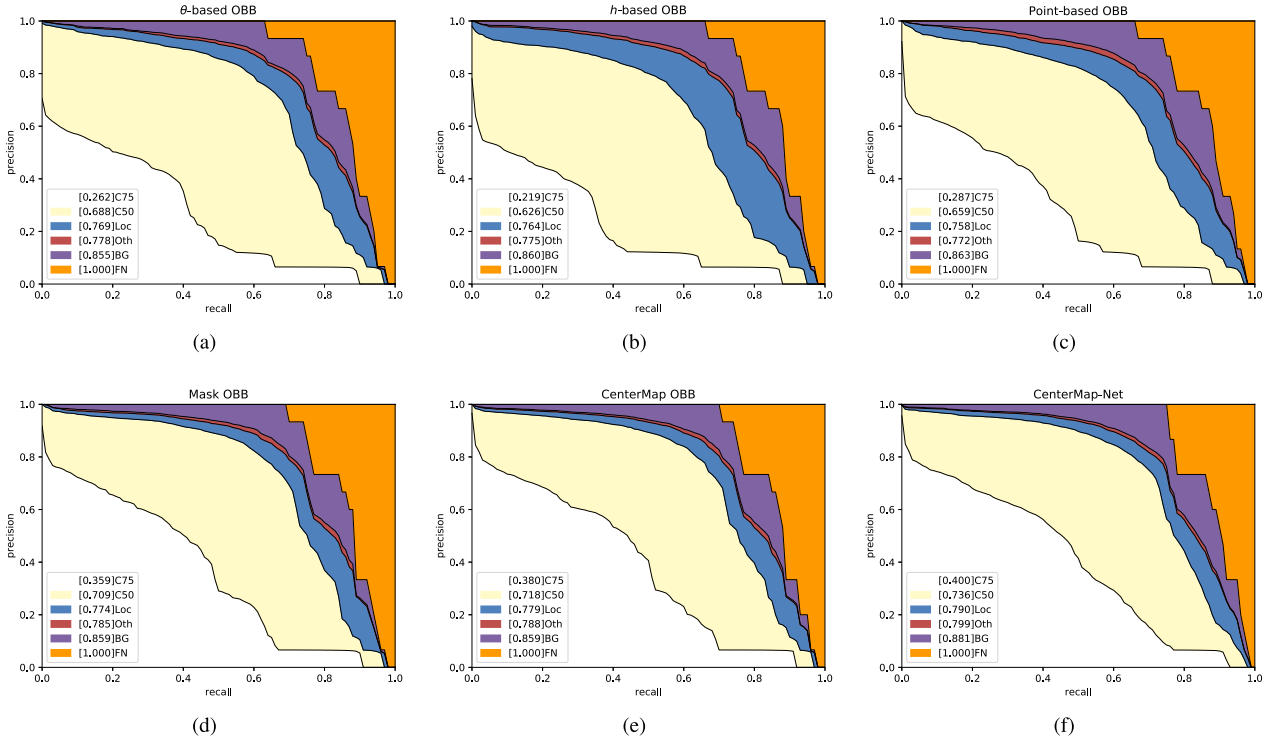


Fig. 9. Diagnosis results on (a)–(e) five different representations and (f) CenterMap-Net in the DOTA validation set. Each plot shows the evolving distribution of FP types as considered. Loc: poor localization; Oth: confused with other object category; and BG: confused with background. Best viewed in color and zoomed-in view. (a) θ-based OBB. (b) h-based OBB. (c) Point-based OBB. (d) Mask OBB. (e) CenterMap OBB. (f) CenterMap-Net.

TABLE VI

COMPARISON WITH MASK OBB AND CENTERMAP OBB ON SELECTED CATEGORIES OVER THE DOTA TEST SET (%). THE BEST RESULT IS HIGHLIGHTED IN BOLD

OBB Representation	PL	BD	SH	ST	RA	HA	HC
Mask OBB	88.20	74.27	83.95	80.70	63.52	63.56	53.51
CenterMap OBB	**88.34**	**78.82**	**84.89**	**80.88**	**64.11**	**64.00**	**56.69**

the legend. We adopt the publicly available detection analysis tool from [70] for these illustrations. The items of analysis are listed as follows.

1) *C75:* PR curve at IoU = 0.75—area under curve corresponds to $AP^{IoU=0.75}$ metric.
2) *C50:* PR curve at IoU = 0.5—area under curve corresponds to $AP^{IoU=0.5}$ metric.
3) *Loc:* PR curve at IoU = 0.1, which ignores the localization errors. Specifically, a detection box that has an overlap (IoU $\in [0.1, 0.5)$]) with any ground-truth box is labeled as localization error. The area between C50 and Loc measures the influence of the localization error.
4) *Oth:* PR curve after all class confusions are removed. Oth is computed by setting all other objects to have the same class label as the class in question.
5) *BG:* PR curve after all background (and class confusion) FPs are removed.
6) *FN:* PR curve after all remaining errors are removed.

According to Fig. 9(a)–(e), the CenterMap OBB can significantly improve the localization ability when IoU = 0.5 (C50) or IoU = 0.75 (C75). For C50, compared with other OBB representations, the CenterMap OBB is up to 9.2% higher. For C75, which uses the strict IoU threshold, the CenterMap OBB can improve the AP from 21.9% to 38.0%, indicating that CenterMap OBB can better match oriented objects in the aerial images. Furthermore, the AP difference between the five OBB representations is less than 2.1% and 0.8% on Loc and BG, respectively, which indicates that the main difference between the five OBB representations is the localization ability, since Loc ignores the localization error and BG ignores the background confusion error. Thus, we find that the performance gain of the CenterMap OBB is mainly due to the improvement of localization ability.

C. Ablation Study

To verify the effectiveness of our method, we conduct a series of comparative experiments on the DOTA test set. Table VII summarizes the results of our models with different settings. The detailed comparison is given in the following.

1) Baseline: The Mask OBB [22] without other components is used as the baseline of the ablation experiments. To ensure the fairness and accuracy, all experimental data and hyperparameters are strictly consistent. We use ResNet-50-FPN as the backbone and mAP as the indicator of model performance.

2) CenterMap OBB: Compared with the baseline that uses the binary segmentation map as the ground truth of the OBB branch, CenterMap OBB uses the CenterMap as the ground

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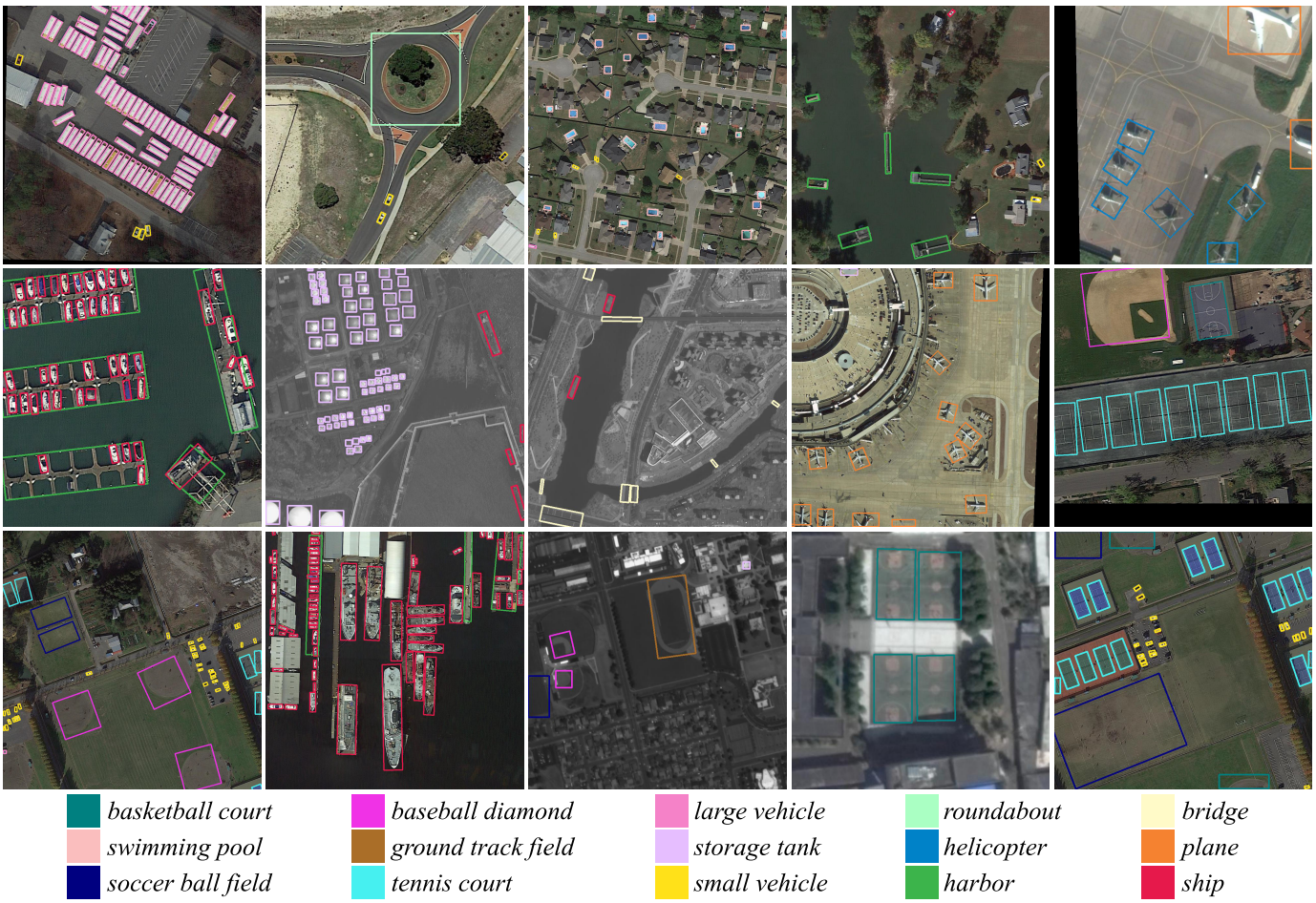


Fig. 10. Visualization of detection results using CenterMap-Net on the OBB task of the DOTA data set. Best viewed in color and zoomed-in view.

TABLE VII
ABLATION STUDY OF THE PROPOSED CENTERMAP-NET OVER THE DOTA TEST SET

Method	OBB (%)	HBB (%)
baseline	69.97	70.14
CenterMap OBB	70.99	70.94
CenterMap OBB + WPSGA-Net w/o Weight Map	71.34	71.37
CenterMap OBB + WPSGA-Net w/ Weight Map (CenterMap-Net)	71.59	71.74

truth. Note that we use the same network structure as baseline in this experiment. It achieves an improvement of 1.02% on the OBB task and 0.80% on the HBB task. This implies that the CenterMap OBB is a better representation for OBB in the aerial images.

3) CenterMap OBB + WPSGA-Net w/o Weight Map: As mentioned in Section III, WPSGA-Net uses the pseudosegmentation and weight map as the ground truth and loss weight, respectively. It can provide object-level features for predicting the HBBs and OBBs. The *CenterMap OBB + WPSGA-Net w/o Weight Map* means we do not use weight map to adjust segmentation loss in the training stage. Table VII shows that this module can achieve 0.35% mAP and 0.43% mAP improvements on the HBB task and the OBB task, demonstrating the effectiveness of our proposed pseudosegmentation-guided attention module.

4) CenterMap OBB + WPSGA-Net w/ Weight Map (CenterMap-Net): Table VII shows that the weight map of WPSGA-Net can further improve the mAP of 0.25% on the OBB task and 0.37% on the HBB task. The results of this ablation study well fit with our motivations.

D. Comparison With the State-of-the-Art Detectors

We compare the performance of our method with the state-of-the-art methods on the OBB task and HBB task of three data sets, i.e., the DOTA [15] data set, HRSC2016 [14] data set, and UCAS-AOD [13] data set.

1) Results on DOTA Data Set: In the final results of DOTA, data augmentation is applied for a fair comparison with other methods such as [18] and [71]. Specifically, we resize the original images at two scales (1.0 and 0.5) before dividing the images into patches; after resizing, we crop the resized images into patches of 1024×1024 pixels with an overlap of 200 pixels in both training and inference stages. In addition, each image is randomly flipped along the horizontal direction with a probability of 0.5 and randomly rotated with an angle selected from $\{0°, 90°, 180°, 270°\}$ in the training stage.

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TABLE VIII

QUANTITATIVE COMPARISON OF THE BASELINES AND OUR METHOD ON THE OBB TASK IN THE TEST SET OF DOTA (%). THE BEST RESULT IN EACH CATEGORY IS HIGHLIGHTED IN BOLD

Method	Backbone	FPN	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP
SSD [4]	VGG-16	-	41.06	24.31	4.55	17.10	15.93	7.72	13.21	39.96	12.05	46.88	9.09	30.82	1.36	3.50	0.00	17.84
YOLOv2 [7]	Darknet-19	-	52.75	24.24	10.60	35.50	14.36	2.41	7.37	51.79	43.98	31.35	22.30	36.68	14.61	22.55	11.89	25.49
R-FCN [5]	ResNet-101	-	39.57	46.13	3.03	38.46	9.10	3.66	7.45	41.97	50.43	66.98	40.34	51.28	11.14	35.59	17.45	30.84
FR-H [3]	ResNet-50	-	49.74	64.22	9.38	56.66	19.18	14.17	9.51	61.61	65.47	57.52	51.36	49.41	20.80	45.84	24.38	39.95
FR-O [15]	ResNet-50	-	79.42	77.13	17.70	64.05	35.30	38.02	37.16	89.41	69.64	59.28	50.30	52.91	47.89	47.40	46.30	54.13
R-DFPN [21]	ResNet-101	✓	80.92	65.82	33.77	58.94	55.77	50.94	54.78	90.33	66.34	68.66	48.73	51.76	55.10	51.32	35.88	57.94
R^2CNN [61]	ResNet-101	-	80.94	65.67	35.34	67.44	59.92	50.91	55.81	90.67	66.92	72.39	55.06	52.23	55.14	53.35	48.22	60.67
RRPN [43]	ResNet-101	-	88.52	71.20	31.66	59.30	51.85	56.19	57.25	90.81	72.84	67.38	56.69	52.84	53.08	51.94	53.58	61.01
ICN [17]	ResNet-101	✓	81.40	74.30	47.70	70.30	64.90	67.80	70.00	90.80	79.10	78.20	53.60	62.90	67.00	64.20	50.20	68.20
RoI Trans. [18]	ResNet-101	✓	88.64	78.52	43.44	75.92	68.81	73.68	83.59	90.74	77.27	81.46	58.39	53.54	62.83	58.93	47.67	69.56
CAD-Net [28]	ResNet-101	✓	87.80	82.40	49.40	73.50	71.10	63.50	76.70	**90.90**	79.20	73.30	48.40	60.90	62.00	67.00	62.20	69.90
SCRDet [59]	ResNet-101	✓	89.98	80.65	52.09	68.36	68.36	60.32	72.41	90.85	**87.94**	**86.86**	**65.02**	66.68	66.25	68.24	65.21	72.61
Li *et al.* [29]	ResNet-101	✓	**90.21**	79.58	45.49	76.41	73.18	68.27	79.56	90.83	83.40	84.68	53.40	65.42	**74.17**	69.69	64.86	73.28
Gliding Vertex [71]	ResNet-101	✓	89.64	**85.00**	52.26	**77.34**	73.01	73.14	86.82	90.74	79.02	86.81	59.55	**70.91**	72.94	70.86	57.32	75.02
CenterMap-Net	ResNet-50	✓	89.02	80.56	49.41	61.98	**77.99**	74.19	83.74	89.44	78.01	83.52	47.64	65.93	63.68	67.07	61.59	71.59
CenterMap-Net*	ResNet-101	✓	89.83	84.41	**54.60**	70.25	77.66	**78.32**	**87.19**	90.66	84.89	85.27	56.46	69.23	74.13	**71.56**	**66.06**	**76.03**

As shown in Tables VIII and IX, we compare our method with the state-of-the-art methods of the OBB and HBB tasks on the DOTA data set. Besides the official baseline given by the DOTA data set, we also compare the proposed method with RRPN [43], R^2CNN [61], R-DFPN [21], ICN [17], RoI Transformer [18], and SCRDet [59]. Note that some methods only report the mAP of the OBB task. By applying ResNet-101 and data augmentation, as shown in Tables VIII and IX, CenterMap-Net* achieves 76.03% and 77.33% mAP on the OBB task and HBB task of the DOTA test set, respectively. Thus, our method obtains the state-of-the-art performance on the DOTA data set. Note that our method attains a smaller gap between its OBB task mAP and HBB task mAP than other methods. Some typical visualization results on the DOTA data set are displayed in Fig. 10.

2) Results on HRSC2016 Data Set: Table X shows the comparison results with the state-of-the-art methods of the HRSC2016 OBB task. Our method achieves the 92.8% mAP of the OBB task and outperforms all other methods of comparison. Note that we do not use the image pyramid scheme and the data-augmentation technology, and the backbone of the network is ResNet-50. Some visualization results on the HRSC2016 data set are displayed in Fig. 11.

3) Results on UCAS-AOD Data Set: Table XI shows the comparison results with the state-of-the-art methods of the UCAS-AOD OBB task. Our full model achieves the 96.26% mAP of the OBB task with the ResNet-50 backbone and outperforms all other methods. For the plane and car categories, our proposed method achieves 97.86% and 94.66% APs, respectively.

E. Detailed Analysis of CenterMap-Net on DOTA Data Set

1) Analysis of WPSGA-Net: As mentioned in Section IV-C, CenterMap-Net includes the CenterMap OBB and the WPSGA-Net. The WPSGA-Net can learn global contexts, aiming at providing object-level features for the detection branch. As shown in Fig. 9(e) and (f), the performance gain is mainly attributed to two factors by using the WPSGA-Net: the

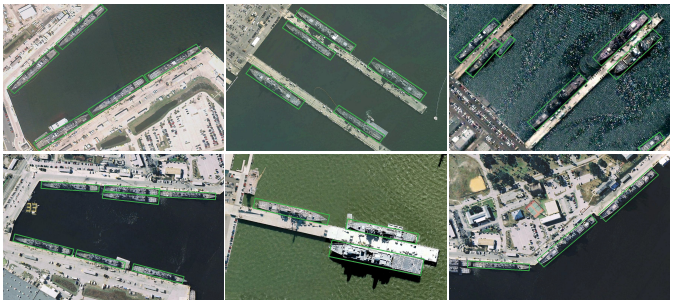
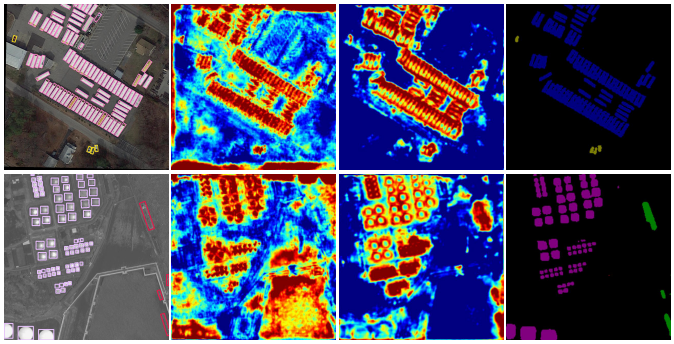


Fig. 11. Visualization of detection results using CenterMap-Net on the HRSC2016 data set.



(a) (b) (c) (d)

Fig. 12. Visualizing the outputs of the WPSGA-Net on DOTA data set. (a) Detection results. (b) Input feature map of attention network. (c) Output feature map of attention network. (d) Segmentation map generated by WPSGA-Net.

improvements of localization ability and classification ability. For localization ability, the localization error is reduced from 6.1% to 5.4% if we compare the gap between C50 and Loc, which measures the localization error. In addition, according to the gap between Loc and FN, the CenterMap-Net can reduce the classification error from 22.1% to 21.0% due to the improvement in classification ability.

Furthermore, the outputs of WPSGA-Net are illustrated in Fig. 12. We observe that the semantic feature map [shown as Fig. 12(c)] and the generated segmentation map [shown

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TABLE IX

QUANTITATIVE COMPARISON OF THE BASELINES AND OUR METHOD ON THE HBB TASK IN THE TEST SET OF DOTA (%). THE BEST RESULT IN EACH CATEGORY IS HIGHLIGHTED IN BOLD

Method	Backbone	FPN	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP
SSD [4]	VGG-16	-	57.85	32.79	16.14	18.67	0.05	36.93	24.74	81.16	25.10	47.47	11.22	31.53	14.12	9.09	0.00	29.86
YOLOv2 [7]	Darknet-19	-	76.90	33.87	22.73	34.88	38.73	32.02	52.37	61.65	48.54	33.91	29.27	36.83	36.44	38.26	11.61	39.20
R-FCN [5]	ResNet-101	-	81.01	58.96	31.64	58.97	49.77	45.04	49.29	68.99	52.07	67.42	41.83	51.44	45.15	53.30	33.89	52.58
FR-H [3]	ResNet-50	-	80.32	77.55	32.86	68.13	53.66	52.49	50.04	90.41	75.05	59.59	57.00	49.81	61.69	56.46	41.85	60.46
ICN [17]	ResNet-101	✓	90.00	77.70	53.40	73.30	73.50	65.00	78.20	90.80	79.10	84.80	57.20	62.10	73.50	70.20	58.10	72.50
SCRDet [59]	ResNet-101	✓	**90.18**	81.88	55.30	73.29	72.09	77.65	78.06	**90.91**	82.44	**86.39**	**64.53**	63.45	75.77	78.21	60.11	75.35
Li *et al.* [29]	ResNet-101	✓	90.15	78.60	51.92	**75.23**	73.60	71.27	81.41	90.85	83.94	84.77	58.91	65.65	76.92	79.36	**68.17**	75.38
CenterMap-Net	ResNet-50	✓	88.88	81.24	53.15	60.65	78.62	66.55	78.10	88.83	77.80	83.61	49.36	66.19	72.10	72.36	58.70	71.74
CenterMap-Net*	ResNet-101	✓	89.70	**84.92**	**59.72**	67.96	**79.16**	**80.66**	**86.61**	90.47	**84.47**	86.19	56.42	**69.00**	**79.33**	**80.53**	64.81	**77.33**

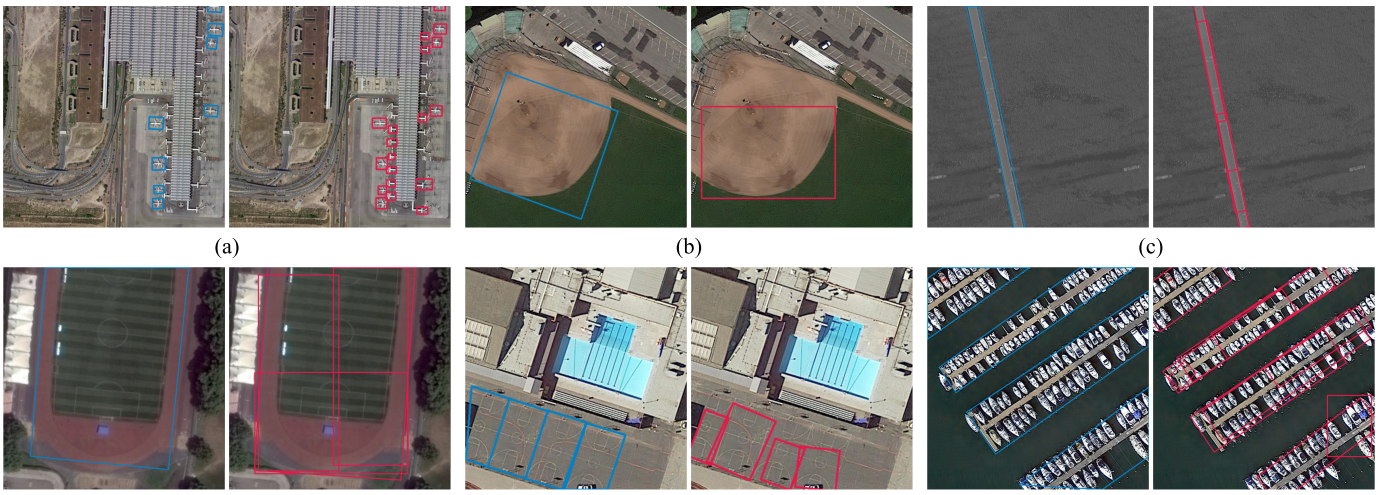


Fig. 13. Some typical failure predictions of CenterMap-Net on the DOTA data set. The left image is annotated with ground truth and the right image is the detection results in each subfigure. (a) Plane. (b) Baseball diamond. (c) Bridge. (d) Ground track field. (e) Basketball court. (f) Harbor.

TABLE X

COMPARISON WITH THE STATE-OF-THE-ART METHODS ON OBB TASK OF HRSC2016. THE BEST RESULT IS HIGHLIGHTED IN BOLD (THE VOC12 AP METRIC IS USED)

Method	mAP (%)
BL2 [42]	69.6
RC1 [42]	75.7
RC2 [42]	75.7
R^2PN [72]	79.6
RRD [19]	84.3
RoI Trans. [18]	86.2
Gliding Vertex [71]	88.2
RBOX-CNN [73]	91.9
CenterMap-Net	**92.8**

TABLE XI

COMPARISON WITH THE STATE-OF-THE-ART METHODS ON UCAS-AOD OBB TASK. THE BEST RESULT IS HIGHLIGHTED IN BOLD (THE VOC12 AP METRIC IS USED)

Method	Plane (%)	Car (%)	mAP (%)
Xia *et al.* [15]	-	-	89.41
ICN [17]	-	-	95.67
CenterMap-Net	**97.86**	**94.66**	**96.26**

as Fig. 12(d)] by the WPSGA-Net fit well with the detection results, indicating that the WPSGA-Net can provide the object-level features by learning global contexts for the detection branch to predict HBBs and OBBs more accurately. In addition, Fig. 12(b) and (c) shows that the attention network can enhance the object information and weaken the nonobject information in the feature map.

2) Limitation of CenterMap-Net: Fig. 13 shows some typical failure predictions of CenterMap-Net. Fig. 13(a) shows that the detection results of the plane are confused with the background that has a similar texture and color. As shown in Fig. 13(b), the baseball diamond may not be detected correctly due to the fuzzy orientation. Furthermore, Fig. 13(c)–(f) shows that the local parts of the bridge, harbor, and ground track field are detected in some situation, since the objects are too large. In addition, some basketball courts may not be detected correctly due to the dense arrangement, as shown in Fig. 13(e).

3) Analysis of Categorywise Performance: In Fig. 14, we apply the analysis tool mentioned in Section IV-B4 to analyze the errors of per category. We observe that most failure detection comes from the BG, which may be due to three main reasons: the first one is that some backgrounds occur frequently with particular categories that fool the detector as foregrounds such as bridge and road (background), the second one is that some small objects are not annotated in the

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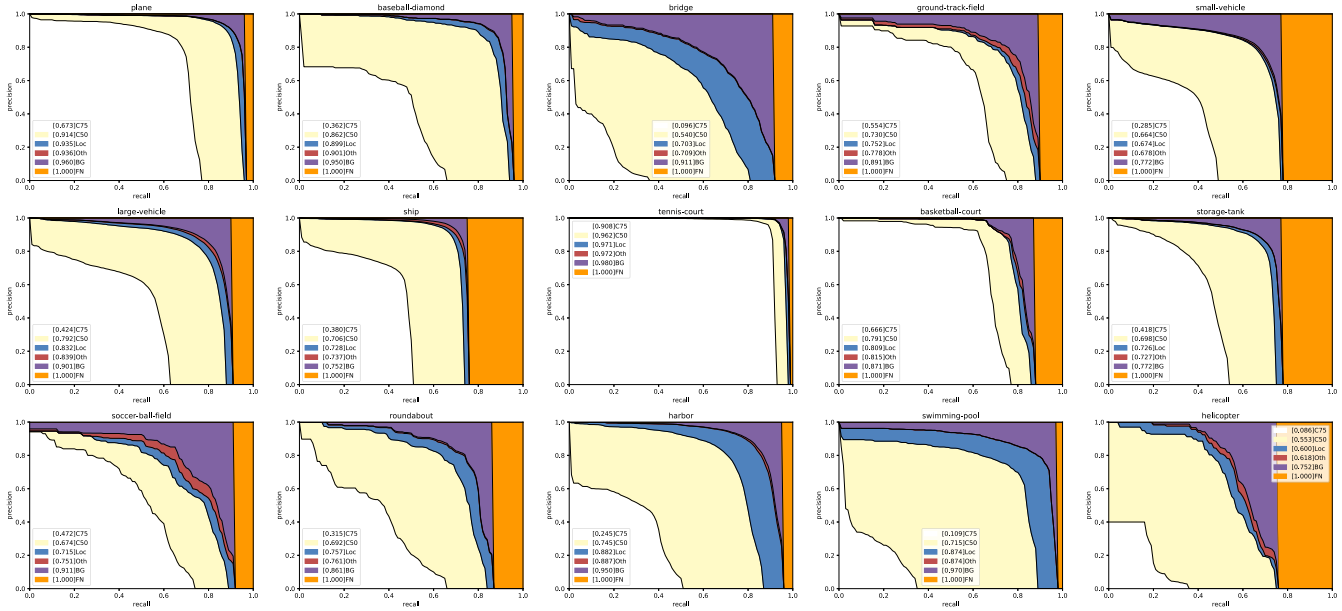


Fig. 14. Detailed analysis of CenterMap-Net on each category in the DOTA validation set. Best viewed in color and zoomed-in view.

DOTA data set such as small vehicle, and the third one is that some categories are rare in the aerial images like helicopter. Specifically, as shown in Fig. 14, the bridge, small vehicle, and helicopter can obtain the improvements of 20.2%, 9.4%, and 13.4% on the AP metric, respectively. In addition, the Loc is also an important reason for failure detection. For instance, removing the Loc can obtain the AP improvements of 16.3%, 13.7%, and 15.8% for the bridge, harbor, and swimming pool, respectively, which is mainly attributed to the bridge and harbor that are very long in aerial images, and the orientation of the swimming pool cannot be well defined.

V. Conclusion

In this article, we analyzed the influence of different representations for the OBB in the aerial images, attempting to overcome the shortcomings of the typical regression-based OBB representations and the Mask OBB representation. Based on that, the CenterMap OBB representation was proposed to tackle the target definition ambiguity of the regression-based OBB representations and the background pixel ambiguity in the general binary segmentation map. With the CenterMap OBB, we further proposed the WPSGA-Net to generate the object-level features, which enhances the features of generating HBBs and OBBs. The experimental results on the DOTA, HRSC2016, and UCAS-AOD data sets demonstrated the superior performance of our proposed CenterMap-Net over the state-of-the-art ones. Notably, our method achieves 76.03% for the OBB task and 77.33% for the HBB task on DOTA. Meanwhile, it achieves 92.8% and 96.26% on the HRSC2016 and UCAS-AOD OBB tasks. Our codes are available at: https://github.com/jwwangchn/aidet.

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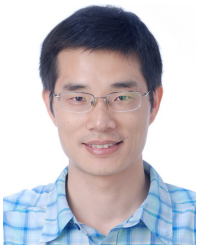
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